Hardware-Oriented SNR-Adaptive Model-Informed LISTA for FPGA-Based Passive-RIS Channel Estimation

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***Abstract*—To solve the problem of increased complexity in channel estimation when utilizing Reconfigurable Intelligent Surfaces (RIS), this work is mainly concentrated on developing a hardware-oriented, model-informed Learned Iterative Shrinkage-Thresholding Algorithm (LISTA) to be implemented on a Field-Programmable Gate Array (FPGA). This 16-layer LISTA core is based upon unrolling the Iterative Shrinkage-Thresholding Algorithm (ISTA), and utilizes Binary Rademacher pilots, an Early Noise Adaptation Network (ENE), and a Residual Convolutional Neural Network (CNN). The LISTA core has a total of 230,749 parameters. For testing purposes, utilized 28 GHz ray-traced DeepMIMO channels ($M/N = 256/1040 \approx 0.246$). With a Signal-to-Noise Ratio (SNR) of 30 dB, the Normalized Mean-Square Error (NMSE) was -14.9 dB. Compared to Least Squares (LS) and ISTA, this represents improvements of 13.7 dB and 9.6 dB respectively. When executing as fixed-point C++ using High-Level Synthesis (HLS), there is a loss of 0.065 dB relative to 32-bit floating-point execution. When executed on the Xilinx ZCU104 (100 MHz), the execution time was 23.27 ms; of the 1,000 held-out vectors, 933 achieved NMSE ≤ -10 dB, with an average NMSE of -16.96 dB. Comparisons with Zynq-based Orthogonal Frequency-Division Multiplexing (OFDM) estimators allowed us to assess hardware-related trade-offs associated with underdetermined compressive recovery.**

***Index Terms*—Channel estimation, compressed sensing, deep learning, field programmable gate arrays, quantization (signal), reconfigurable intelligent surfaces.**

I. Introduction

THE mmWave signal degrades quickly. It is blocked frequently by walls, terrain, and pedestrians. RIS can assist in keeping the link intact by reflecting the transmitted signals of these types of obstacles [1]. RIS is a flat surface consisting of small metal or plastic pieces that are controlled with PIN diodes or varactors. Therefore, it requires very little static energy and does not require expensive RF components [2]. The disadvantage of being simple is that there are no active transceivers and no ADCs. The surface cannot decode, correlate, or produce pilots. Consider 64 elements on the surface and 16 antennas at the base station; this results in 1040 unknown parameters for the cascaded channel. Least squares estimation would require at least 1040 pilot slots, which is much larger than what is available within a typical coherence interval.

Sparse recovery can be the answer to the pilot overhead problem. Millimeter wave (mmWave) propagation is characterized by a small number of dominant specular reflections, which means the cascaded channel can be represented sparsely in the virtual angular domain. Although these estimators perform well, they typically require hundreds of iterations to converge, making them computationally intensive for real-time baseband processing. Deep Unfolding solves the computational complexity issue associated with iterative optimization algorithms by transforming an iterative optimization algorithm into a feedforward network of fixed length [19]. A classic example is Learned ISTA (LISTA) [14], which is a version of Learned Iterative Shrinkage-Thresholding Algorithm (LISTA) that is unrolled into 10–20 layers with learned filter matrices and thresholds. Chen et al. demonstrated that partial weight coupled LISTA exhibits linear convergence characteristics [15]. Other similar approaches include Analytic Filters in ALISTA [16], Hyperparameter Tuning [17], Neural Adaptation [18], and Message-Passing Architectures such as AMP, VAMP, and LAMP [7], [8], [21]–[23]. Additionally, Residual CNN denoisers can be used to remove residual estimation errors [6].

Moving these ideas onto real hardware exposes three gaps in the literature:

1) *No relationship with hardware realization:* Most RIS estimators developed to date have been evaluated primarily through either Python or MATLAB simulation. The number of slices used, block RAMs required, DSP utilization and clock latency have been rarely reported.
2) *Coherence issues with DFT pilots:* Many authors typically use truncated Discrete Fourier Transform (DFT) matrices for the RIS phase shifts. In situations where there are less observations than unknown parameters ($M \ll N$), these types of matrices will exhibit very high mutual coherence and therefore cause problems in recovering a sparse representation of the signal.

3) *No On-Board Validation of Sparse Solvers:* However, interpolating OFDM pilots over statistical channels represents a different problem than compressed sensing of cascaded RIS channels using ray-traced data representing real-world conditions.

To bridge the gap between algorithm and silicon, developed a co-designed unrolled estimator called "Model-Informed LISTA" and tested its performance on an AMD/Xilinx ZCU104 board. this work contributions include the following:

1) *Rademacher Pilots* use binary Rademacher phase sequences (± 1) to drive the surface, rather than complex DFT codes. For this work underdetermined setup, where there are 256 complex measurements and 1,040 unknowns ($M/N \approx 0.246$), using Rademacher phase sequences reduces the mutual coherence of the system from 0.984 to 0.603. Additionally, Rademacher phase sequences result in an improvement of software NMSE of approximately 1.87 dB. Reflection states can only take on values of 0 or π. Therefore, the DSP multiplier required to compute matrix projections on-chip is eliminated and the computation can be performed utilizing only LUT-based adder trees.
2) *An unrolled core based on models, adaptive to SNR:* unrolled 16 iterations of ISTA using the same weight matrix for all of the iterations, this allows us to keep the amount of data stored in parameters below 1 MB. An early-stage noise estimation network (ENE) reads an estimate of the noise level directly from the measurement signal received. This ENE then adjusts the shrinkage thresholds. As a result, one bit stream will cover a wide range of SNR levels from -10 to +30 dB. After removed most of the reconstruction artifact, use a very compact two-layer residual CNN to remove any remaining artifact.
3) *Fixed-point C++ HLS realization:* create custom pipeline kernels using Vitis HLS 2022.2. A word-length study has shown us that 16-bit fixed point (`ap_fixed`) lost only 0.065 dB when compared to 32-bit floating point, making it a realistic point of trade-off between precision and resthis workces.
4) *Validation on 1000 vectors using a measured board:* A 100 MHz bitstream was created for the ZCU104 and then tested with 1000 previously unused deep MIMO ray-traced channels. On average, the hardware had a -16.96 dB NMSE and 93.3
5) *System Bottleneck Analysis:* assess the memory and scheduling overhead across the ARM Cortex-A53 and the FPGA fabric. In addition, record how end-to-end execution is affected by AXI DMA resets, cache flushes, and lat transpositions.

The next part of this paper will be structured as follows, section II discusses other similar papers published. Sections III through IV explain the signal model, how the model is used in LISTA's pipeline, and the baseline solvers used for comparison. Section V explains the hardware used for testing and its FPGA datapath. The experimental results and board measurements are discussed in sections VI to IX. At last, section X analyzes the hardware trade-offs and section XI concludes.

II. Related Work

RIS channel estimation: DS-OMP uses the fact that the user's angular support is similar to that of the BS's channel. As the array size increases, the number of matrix calculations required by DS-OMP increases [4]. Distributed learning was employed to estimate channels at different nodes [5]. Deep convolutional denoisers re-utilized to improve the accuracy of least squares estimates of channels [6]. Recent studies have also tuned RIS reflection patterns [12], developed self-supervised methods for learning channel representations [11], and employed attention networks to compress CSI data [13]. All of these studies assessed their proposed algorithms through simulation only. They did not consider the memory footprints of their algorithms, fixed point quantization effects, or the impact of hardware timing on algorithm performance.

Deep unrolling and message-passing: Algorithm unrolling replaces manual parameter selection of iterative solvers with gradient-based training. Gregor LeCun are the first to unroll ISTA into LISTA for sparse coding [14]; Chen demonstrated that partial weight coupled LISTA has linear convergence [15]. Other versions make it easier to train. ALISTA calculates its projection weights analytically using pseudo-inverse matrices [16]. HyperLISTA selects only three global scaling parameters [17]. Neurally augmented ALISTA chooses a step size for each iteration [18]. Message-passing algorithms are also unrolled. Approximate message-passing (AMP) [21] is asymptotically optimal for independent and identically distributed (i.i.d.) Gaussian dictionaries; Vector AMP (VAMP) [22] extends that to right-orthogonal invariant matrices; LAMP [23] learns the linear filter coefficients. RIS-specific unrolled versions include learned VAMP [7], deep AMP [9], multi-block attention [8], and dual-order ISTA [10]. To this work knowledge, no paper has reported physical synthesis results or board level testing.

FPGA Implementation of Learned Estimators: Hardware implementation of deep learning based channel estimation is relatively rare. Haq et al. used a Zynq MPSoC to implement a deep neural network for OFDM preamble-based estimation [28]. Post-training quantization of learned estimators often requires longer word lengths than quantization-aware training [29]. To the best of our knowledge, no one has yet generated a fixed-point unrolled network for underdetermined RIS channel recovery on an FPGA and evaluated its performance.

III. System Model

A. Measurement Model

Evaluation on ray-traced channels from the 28 GHz DeepMIMO O1_28 outdoor scenario [27]. Let $\mathbf{h} \in \mathbb{C}^N$ be the vectorized cascaded BS–RIS–user channel. The base station has $N_{\mathrm{BS}} = 16$ antennas and the surface has $N_{\mathrm{RIS}} = 64$ reflecting elements. One direct user-to-BS path is added, which gives $N = 16 \times (64+1) = 1040$ complex channel coefficients. During

TABLE I
POSITIONING AGAINST REPRESENTATIVE WORK ("–" MEANS NOT REPORTED IN THE CITED WORK, TO THE BEST OF THE AUTHORS' KNOWLEDGE)

Method	RIS	Sparse/ unfolded	Pilot matrix	Fixed-pt. / Quant.	On-board measured
DS-OMP [4]	✓	✓	–	–	–
CNN denoiser [6]	✓	–	–	–	–
Learned VAMP [7]	✓	✓	–	–	–
Multi-block attention [8]	✓	✓	–	–	–
Model-Informed LISTA	✓	✓	DFT vs. Rademacher	Post-train conversion	✓

training, a single-antenna user sends pilots over $T = 16$ time slots while the surface steps through 16 phase patterns. The baseband signal recorded at the base station follows

$$\mathbf{y} = \Phi\mathbf{h} + \mathbf{n}, \quad \Phi \in \mathbb{C}^{M\times N},\, M \ll N, \tag{1}$$

where $M = 256$ complex measurements (16 slots $\times 16$ antennas) and $\mathbf{n} \sim \mathscr{CN}(\mathbf{0}, \sigma^2\mathbf{I})$ is additive white Gaussian noise.

The direct link bypasses the surface, while the 64 reflected paths are modulated by the RIS elements. The sensing matrix Φ therefore has a block-diagonal Kronecker structure:

$$\Phi = V \otimes \mathbf{I}_{16}, \tag{2}$$

with the composite phase schedule $V \in \mathbb{R}^{16\times 65}$ split as

$$V = [V_{\text{RIS}} \;\; \mathbf{v}_{\text{direct}}]. \tag{3}$$

Here $\mathbf{v}_{\text{direct}} = \mathbf{1}_{16}/\sqrt{16}$ stands for the unmodulated direct path, and $V_{\text{RIS}} \in \mathbb{R}^{16\times 64}$ holds the time-varying reflection states. gather $M = 256$ observations for $N = 1040$ channel parameters, so the sensing ratio is $M/N = 256/1040 \approx 0.246$. The operating SNR is $\text{SNR} = \mathbb{E}[\|\Phi\mathbf{h}\|_2^2]/\mathbb{E}[\|\mathbf{n}\|_2^2]$, and sep it from -10 to $+30$ dB.

B. Sparsity and Rademacher Sketching

Millimeter-wave channels concentrate their energy along a few spatial directions. A 2D spatial discrete fourier transform $\mathbf{F}_{\text{RIS}} \in \mathbb{C}^{64\times 64}$ on the RIS elements, together with the identity on the direct path, gives the block transform $\widetilde{\mathbf{F}} = \text{diag}(\mathbf{F}_{\text{RIS}}, \mathbf{I}_{16}) \in \mathbb{C}^{1040\times 1040}$. In this virtual angular domain the transformed channel $\mathbf{x} = \widetilde{\mathbf{F}}\mathbf{h}$ is sparse, with only about 6% non-zero entries. The measurement model becomes $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{n}$ with dictionary $\mathbf{A} = \Phi\widetilde{\mathbf{F}}^H$, and sparse recovery reduces to the LASSO problem

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \tfrac{1}{2}\|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2 + \lambda\left(\|\Re(\mathbf{x})\|_1 + \|\Im(\mathbf{x})\|_1\right). \tag{4}$$

The sparse recovery success depends on conditioning of sensing matrix $\mathbf{A}$. Compressive sensing theory tells us that random Bernoulli or Rademacher matrices satisfy the Restricted Isometry Property (RIP) with high probability [26]. use this fact directly. The physical RIS states are set to binary 0 or π phase shifts, so the entries of V_{RIS} become $\pm 1/\sqrt{16}$. Table II shows that this Rademacher design beats a standard DFT phase matrix, which form from 16 rows of a normalized 65×65 DFT matrix. The mutual coherence $\mu(\mathbf{A}) = \max_{i\neq j} |\mathbf{a}_i^H\mathbf{a}_j|/(\|\mathbf{a}_i\|_2\|\mathbf{a}_j\|_2)$ falls from 0.984 to 0.603.

TABLE II
SENSING-MATRIX QUALITY STATISTICS: DFT VERSUS RADEMACHER PILOT PATTERNS

Statistic	DFT pattern	Rademacher
Mutual coherence $\mu(\mathbf{A})$	0.984	0.603
Nonzero singular-value spread $\sigma_{\max}/\sigma_{\min,\text{nz}}$ of Φ	1.000	2.652
Empirical RIP proxy δ_s^{emp} $(s = 65)$	3.063	1.957

The binary choice also helps the hardware. V becomes purely real. Multiplying by Φ and by $\Phi^H = \Phi^T$ then reduces to sign flips and additions scaled by $1/\sqrt{16}$. absorb that scale factor into the step size of the unrolled network, so the FPGA accelerator computes the sensing projections with plain adder trees and uses no DSP multiplier blocks for them.

C. Metrics

Estimation accuracy is measured by the normalized mean-square error in decibels,

$$\text{NMSE} = 10\log_{10}\left(\frac{\|\hat{\mathbf{h}} - \mathbf{h}\|_2^2}{\|\mathbf{h}\|_2^2}\right), \tag{5}$$

computed over all $2N = 2080$ real and imaginary channel entries. For hardware validation report both the dB-domain average and the linear-domain mean. Across the 10 initial verification vectors, the mean NMSE is -18.73 dB (-15.90 dB in linear scale). Over the full 1000-vector on-board campaign, the linear-domain average is -14.44 dB.

D. Measurement Model

The BS has $N\text{BS} = 16$ antennas and the RIS has $N_{\text{RIS}} = 64$ reflecting elements. A single direct user-to-BS path is included and therefore there are $N = 16 \times (64 + 1) = 1040$ complex channel coefficients. For the purpose of training, a single-antenna user transmits pilots during $T = 16$ time intervals while the RIS changes its phase pattern 16 times.

Standard LISTA, however, replaces the sensing matrix with full dense matrices. A 16 layer model using dense 256×1040 matrices at this work dimension would require $16 \times 256 \times 1040 = 4,259,840$ parameters which is over 4.2 million parameters. This quantity will exceed the capacity of the on-chip BRAM and overwhelm the memory interface of an embedded SoC.

In this work Model-Informed LISTA is a much smaller model because it retains the physical sensing matrix Φ and the virtual domain transformation $\widetilde{\mathbf{F}}$ in each layer of the network. Only the scalar step-sizes γ_k and threshold values θ_k will be learned. The initial input to this work first layer will be the matched-filter adjoint $\mathbf{h}^{(0)} = \Phi^H\mathbf{y}$ that requires no

Algorithm 1 Model-Informed LISTA inference

Require: $\mathbf{y}$, Φ, $\widetilde{\mathbf{F}}$, learned $\{\gamma_k, \bar{\theta}_k\}_{k=0}^{K-1}$, ENE, CNN

$\mathbf{h}^{(0)} \leftarrow \Phi^H \mathbf{y}$

$\mathbf{s} \leftarrow \mathrm{ENE}(\mathbf{y})$

for $k = 0$ to $K-1$ **do**

  $\mathbf{z} \leftarrow \mathbf{h}^{(k)} + \gamma_k \Phi^H \left(\mathbf{y} - \Phi \mathbf{h}^{(k)}\right)$

  $\mathbf{h}^{(k+1)} \leftarrow \widetilde{\mathbf{F}}^H \eta_{\mathbf{s}\cdot\bar{\theta}_k} \left(\widetilde{\mathbf{F}}\mathbf{z}\right)$

end for

$\hat{\mathbf{h}} \leftarrow \mathbf{h}^{(K)} + \mathrm{CNN}\left(\mathbf{h}^{(K)}\right)$

return $\hat{\mathbf{h}}$

multiplication. For each of the layers $k = 0, \ldots, K-1$, it will compute

$$\mathbf{z}^{(k)} = \mathbf{h}^{(k)} + \gamma_k \Phi^H \left(\mathbf{y} - \Phi \mathbf{h}^{(k)}\right), \tag{6}$$

$$\mathbf{h}^{(k+1)} = \widetilde{\mathbf{F}}^H \eta_{\theta_k} \left(\widetilde{\mathbf{F}}\mathbf{z}^{(k)}\right), \tag{7}$$

where η_{θ_k} thresholds the real and imaginary parts independently: $\eta_{\theta_k}(|\cdot|_{\mathrm{real}}) \parallel \eta_{\theta_k}(|\cdot|_{\mathrm{imag}})$.

The use of a single matrix operator on all $K = 16$ layers reduces the number of trainable parameters in the core to 32, i.e., 16 step-size parameters and 16 base-threshold parameters. This will allow the model to remain consistent with the convergence properties of unrolled models that use partial weight coupling (as developed by Chen *et al.* in [15]). The results shows that increasing the number of layers from $K = 12$ to $K = 16$ resulted in an additional improvement of 0.70 dB in NMSE. Increasing the number of layers past $K = 16$ did not result in further improvements.

E. SNR Conditioning and Residual Refinement

The optimal ISTA threshold θ_k will depend upon the level of noise variance (σ^2). A threshold developed for high signal-to-noise ratio (SNR) will allow too much noise to pass through if the channel is poor. Conversely, a threshold developed for low SNR will do the exact opposite and remove physical paths when working under conditions where there is high SNR.

An early noise adaptation network (ENE) will resolve this problem. The ENE, shown in Fig. 1, consists of two fully connected layers (512 inputs to 32 outputs, and 32 inputs to 16 outputs) that convert the input measurement vector $\mathbf{y}$ into an adaptation vector $\mathbf{s} \in \mathbb{R}^{16}$. After converting the measurement vector to the adaptation vector, each layer's activation threshold is adjusted based on the adaptation vector:

$$\theta_k = s_k \bar{\theta}_k, \tag{8}$$

where $\bar{\theta}_k$ represents the learned threshold for each channel. This is a very basic process and allows a single bitstream to accommodate channels with an SNR range of -10 to $+30$ dB, and therefore eliminates the need to reload the weight values into the flash memory.

A compact two layer CNN (kernel size 3, 48 channels) is applied after the sixteen unrolled layers to refine the output: $\hat{\mathbf{h}} = \mathbf{h}^{(K)} + \mathrm{CNN}(\mathbf{h}^{(K)})$. The residual skip connection provides a performance improvement of $+0.96$dB. The reason for this is that the CNN only needs to correct high frequency errors and does not need to recreate the entire channel. The complete pipeline contains 230,749 parameters that consume 0.92MB of memory when stored as 32 bit floating point numbers. train the pipeline end-to-end using Adam with a learning rate of 10^{-3}, a batch size of 64 and an MSE loss function for 100 epochs.

Complexity count: If $\Phi\mathbf{h}$ and $\Phi^H\mathbf{e}$ are calculated using dense math, each layer requires $4MN = 1{,}064{,}960 \approx 1.06 \times 10^6$ real MACs, which means approximately 17.0×10^6 MACs for 16 layers. Because the binary Rademacher matrix is used, all of those products become either addition or subtraction. Therefore, the DSPs that are to be used for those products will be available for the 2D FFT and the residual CNN.

TABLE III
BASELINE ESTIMATORS, OBSERVED BEHAVIOR AND CONFIGURATION

Method	Ref.	Behavior on this problem
LS	[3]	Underdetermined for $M < N$; the minimum-norm pseudoinverse solution does not exploit sparsity; plateau near -1 dB
ISTA	[20]	Fixed threshold; best iterative baseline (≈ -5 dB)
AMP	[21]	Diverges under the proposed structured operator in the experiments
LAMP	[23]	Learned transforms absorb part of the mismatch (≈ -4 dB)
OMP	[24]	Needs sparsity level; coherence-sensitive; plateau near -1 dB
CoSaMP	[25]	Needs sparsity level; ≈ -4 dB at high SNR, poor at low SNR
SVD	[3]	full pseudoinverse ($k = 256$) without any sparsity prior; ≈ -3 dB
VAMP-style	[7], [22]	Simplified variant; near LS (≈ -1 dB)
LMMSE-style	[6]	Sample-covariance linear reference, not a strict bound

IV. BASELINE ESTIMATORS

Comparing our work Model-Informed LISTA estimator to eight baseline estimators and one sample-covariance linear MMSE (LMMSE) reference. To make sure that all of the comparisons are fair, each method receives the same measurement budget, pilot matrices and dimension of matched observations. The baseline estimators are described in Table III as well as what each does when applied to this work problem. Hyperparameters for each of the baseline estimators are listed in Table IV.

Standard LS uses the minimum norm pseudo inverse of (1). The system is under-determined ($M/N \approx 0.246$), therefore LS cannot determine components in the null space of Φ. It has an error floor at approximately -1 dB for all SNRs. OMP and CoSaMP are dependent upon the coherence of the sensing matrix. Both struggled to work with the Kronecker pilot structure, even though they are tuned to the assumed sparsity level $s = 65$.

ISTA [20] is the best of the classical approaches, with an average performance of -5.3 dB when using a SNR of

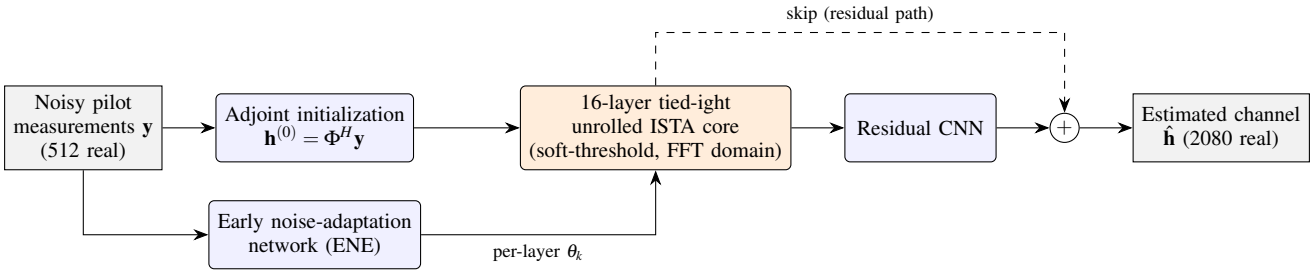


FIG. 1. Model-Informed LISTA pipeline. The CNN refines the unrolled-core estimate, and its output is added back through a residual connection.

TABLE IV
BASELINE CONFIGURATION (REPRODUCIBILITY)

Method	Iterations / stopping	Hyperparameters	Selection
LS	closed form	none	—
ISTA	≤ 500 it.; stop: rel. change $< 10^{-4}$	step=0.5, $\lambda = 0.01$	validation set
AMP	diverged in experiments	Donoho-Maleki-Montanari (DMM)	validation set
LAMP	16 layers	learned	training set
OMP	$s = 65$ (max channels)	none	—
CoSaMP	$s = 65$ (max channels)	none	—
SVD	closed form	full pseudoinverse (k=256)	validation set
VAMP-style	≤ 500 it.	damping: 0.8	validation set
LMMSE-style	closed form	sample covariance	training set

30 dB. AMP [21] was found to diverge in all of this work simulations. The cause of this divergence is readily apparent. AMP's Onsager correction assumes that the measurement matrix ensemble used in the simulation is either independent Gaussian, or right-orthogonally invariant. However, the block diagonal Kronecker structure $\Phi = V \otimes \mathbf{I}_{16}$ violates both of these conditions and creates correlated measurement noise, causing the state-evolution recursion to diverge. On the other hand, Learned AMP (LAMP) [23] can absorb some of the mismatch created by using non-random measurement matrices by training its weight matrices and therefore can reach -4.0 dB. Unlike AMP, ISTA-based methods are independent of the random matrix ensemble and remain stable as long as the step size meets the condition $\gamma_k \leq 2/\|\mathbf{A}\|_2^2$.

V. FIXED-POINT DESIGN AND HARDWARE ARCHITECTURE

A. Design Flow and Quantization

build the accelerator in three steps.

1) *Floating point reference:* A relative error of 1.7×10^{-6} was achieved with this work NumPy based reference implementation, which is bit-accurate to the PyTorch implementation.
2) *C++ HLS description:* created modular C++ sthis workce files for the unrolled LISTA core and the residual CNN. Therefore, developed all of the kernel code in C++ using Vivado HLS directives.
3) *Synthesis and Bitstream Generation:* After that, they are incorporated into a Vivado block design operating at 100 MHz on the AMD Zynq UltraScale+ ZCU104 kit. A bare-metal program running on the Cortex-A53 was used to move data via AXI Direct Memory Access (DMA).

The datapaths use Xilinx fixed-point types: `ap_fixed<16,8>` for weights and activations, and `ap_fixed<32,16>` for accumulators so that nothing overflows. Rounding to a step size Δ adds a variance of $\sigma_r^2 = \Delta^2/12$. Each unrolled layer is non-expansive ($\gamma_k \leq 2/\|\mathbf{A}\|_2^2$), so roundoff accumulates at most linearly with depth. Across the PS–PL boundary, data travels as 32-bit words scaled by 2^{16}.

B. System Partition, Scheduling and Data Movement

Fig. 2 shows the system dataflow. The Cortex-A53 manages the DDR buffers and drives five AXI DMA instances (`dma_lista_y_out`, `dma_lista_init`, `dma_cnn_refine_out`, `dma_cnn_refine2` and `dma_cnn_res`).

Each channel estimate passes through these steps:

The CPU transmits 512 measurement data points ($\mathbf{y}$) and 2080 adjoint initialization data points ($\Phi^H \mathbf{y}$) to the unrolled core using the AXI DMA.

1) The unrolled core is used for running each of the 16 layers and produces a 2080 word intermediate channel estimate.
2) The CPU clears its cache, changes the arrangement of data from a $[16 \times 65 \times 2]$ lat to a channel-first format and sends three separate MM2S streams to the CNN accelerator.
3) At the end of its inference cycle, the CNN asserts `ap_done` and sends the last 2080 words of the channel estimation back to DDR.

The current design will require these processor round trip transactions because there are no on-chip AXI-Stream interconnections between the two IP blocks, the matrix transposition is done by a software routine running on the ARM core, and the AXI DMA driver must reset channels to clear `TLAST` framing states. The estimated transaction transfers 50.8KiB with an effective transfer rate of 2.2MB/s based on the

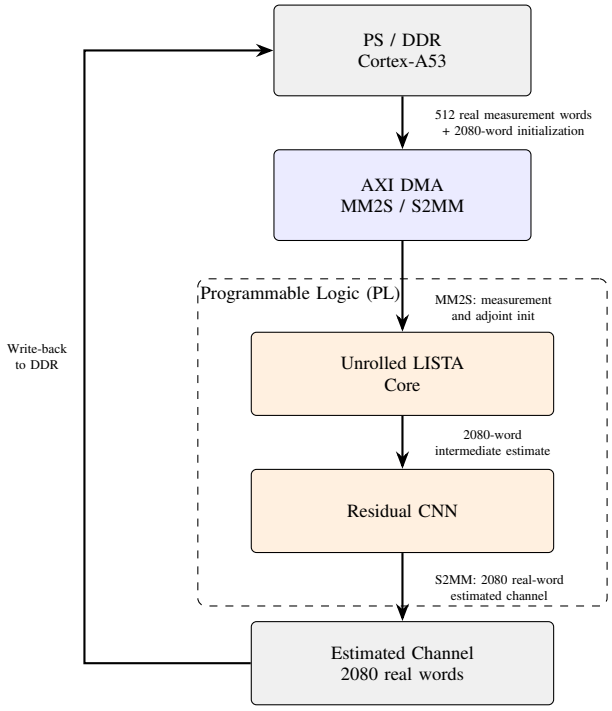


FIG. 2. PS/PL dataflow for channel estimation, showing CPU control, DMA transfers, and sequential LISTA and CNN processing.

estimated transaction time of 23.27 ms. Therefore, the memory bus is not the limiting factor. The total latency is established through software driver handshaking, invalidation of cache and sequential kernel execution.

C. Internal Architecture of the Unrolled Core

In Fig. 3, demonstrates how each unrolled layer of the network is implemented. Each unrolled layer accepts the current state $\mathbf{h}^{(k)}$, calculates the error residual $\mathbf{y} - \Phi\mathbf{h}^{(k)}$. Since Φ contains only $\pm 1/\sqrt{16}$, this step can be compleated using only adders, no DSP and multipliers are required . After calculating the residual, it is multiplied by Φ^H, scaled by γ_k and finally added to provide $\mathbf{z}^{(k)}$.

Next, a two-dimensional (16×64) radix-2 Fast Fourier Transform (FFT) moves the vector into the virtual angular domain. In this domain, the vector then soft-thresholded using the θ_k provided by the ENE, and a two-dimensional Inverse Fast Fourier Transform (IFFT) converts the vector from the angular domain to the spatial domain. The intermediate vectors are stored in dual-port BRAM buffer partitions. Table V includes the synthesis directives.

D. Verification

Verification runs in three phases.

1) *C-simulation:* evaluated the work using a fixed point C++ code against the float32 "golden" model, using a wide range of channel vector types.
2) *C/RTL Co-Simulation:* A total of 20 test cases that represent this work designs are executed using cycle-accurate RTL Co-Simulation. Ten of these cases are evaluated for correctness compared to the Golden Model with a passing criterion of NMSE $<$ -25dB. The remaining ten cases are evaluated for accuracy against Ground Truth with a passing criterion of NMSE $<$ -10dB. All 20 cases passed both criteria.
3) *On-board Execution:* Two tests were conducted on the physical ZCU104.
   - *Campaign V1:* The direct comparison of hardware outputs to the floating-point golden models (pass condition NMSE < -25 dB) provides an isolation of the fixed-point implementation and fidelity of the design.
   - *Campaign V2:* The scoring of the hardware output will be based on how it compares to the actual ray-traced channel (the pass criteria being that the NMSE is less than or equal to -10 dB). This will begin with a total of ten benchmark vectors and will eventually expand to include a total of 1000 held-out channels.

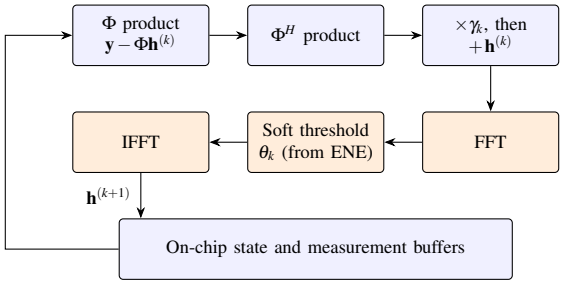


FIG. 3. Datapath of one layer of the unrolled core (blue: sensing-operator and state blocks; orange: transform-domain blocks).

TABLE V
IMPLEMENTATION PARAMETERS OF THE HLS CORES

Item	Value
Clock / HLS target	100 MHz (timing met post-route)
Number of PEs, parallelism factor	16 parallel accumulators, unroll factor 16
Loop unroll / pipeline II (core)	PIPELINE II=1
Loop unroll / pipeline II (CNN)	PIPELINE II=1
Array partitioning, BRAM ports	Complete partitioning (16 blocks), dual-port BRAM
Arrays resident on-chip	weights, thresholds, intermediate states, measurements
FFT: length, precision, architecture	2D Radix-2 16x64 Burst I/O, `ap_fixed<16,8>`
Stream interfaces	AXI-Stream, AXI DMA

VI. EXPERIMENTAL SETUP

Channel dataset: The deepMIMO O1_28 urban outdoor ray tracing dataset [27] was used as work evaluation of propagation path. The base station (BS) contains 16 uniform linear antenna array, and the RIS is composed of 64 planar element. From the initial 100 user grid rows, randomly selected 100,000 different

TABLE VI
EXPERIMENTAL CONFIGURATION (REPRODUCIBILITY)

Item	Value
Dataset	DeepMIMO O1_28 (28 GHz), BS 1, first 100 user-grid rows; $N_{\text{users}} = 100{,}000$ unique spatial channel geometries (1,000 users per each of 100 spatial grid rows, 1 noise realization per channel). The dataset is split strictly geographically so no user geometry overlaps between train and test.
RIS / antennas	predefined RIS position; dimensions: $M = 256, N = 1040$; 16 BS antennas, 64 RIS elements
Split	80/10/10%; 100,000 unique channel geometries
Normalization	Unit variance per channel
Training	Adam, 10^{-3}, batch 64, 100 epochs; MSE loss, seed 0, best validation loss, patience 10
ENE	2 FC layers (512-32-16), ReLU/Sigmoid, yields threshold scales
Hardware	ZCU104, 100 MHz, Vitis/Vivado 2022.2, AXI DMA; pragmas: Table V
Power	see Section VIII
Baselines	Table IV

user positions. This split is based upon geography (80% train, 10% validation, 10% test); therefore, there are no overlapping coordinates between any user position in the test set and the training pool. Therefore, this split allows us to generalize over space,later normalized each channel vector to have a mean energy equal to one. The configuration parameters for the simulation are provided in Table VI.

Software study: One trained Model-Informed LISTA model was evaluated at each SNR (Signal to Noise Ratio) value, ranges from -10 dB to +30 dB with 5 dB increments. The ENE(Early Noise Estimator) adjusts the threshold values depending on the SNR.

Study of the Hardware: The post-route bitstream will be programmed into the ZCU104 which is operating at 100 MHz. A single bare-metal C application running on the Cortex-A53 is used to run the test vectors and stream log data via UART at a rate of 115,200 baud. Latency measurements are obtained through on-board hardware timers.

VII. SOFTWARE RESULTS

A. NMSE versus SNR

Figure 4 and Table VII shows **NMSE over the SNR range from -10 to $+30$ dB.** The Model-Informed LISTA algorithm outperforms all eight comparative algorithms with a large difference. Specifically, at 0 dB SNR the Model-Informed LISTA algorithm provides -13.34 dB NMSE, which g ives a difference of 11.0 dB lesser than the NMSE provided by ISTA and 7.5 dB ,and also lesser than the NMSE provided by the sample-covariance linear reference algorithm. In addition to that, at 30 dB SNR the Model-Informed LISTA algorithm provides -14.93 dB NMSE, which is 9.6 dB lesser than the NMSE provided by ISTA and 13.7 dB lessar than the NMSE provided by Least Squares. Additionally, although the Least Squares and CoSaMP algorithms provide positive NMSE values ($> +8$ dB) at -10 dB SNR, the Model-Informed LISTA algorithm remains capable of providing recovery near -8 dB at -10 dB SNR.

TABLE VII
EXACT NMSE (dB) FROM EXPERIMENT LOGS

Estimator	0 dB	10 dB	20 dB	30 dB
LS	-0.51	-1.20	-1.21	-1.21
VAMP-style	-0.53	-1.35	-1.42	-1.43
CoSaMP	$+0.54$	-3.31	-4.12	-4.21
SVD	-2.82	-3.32	-3.35	-3.35
LAMP	-3.71	-3.65	-3.82	-4.01
ISTA	-2.31	-4.85	-5.32	-5.34
LMMSE-style	-5.81	-11.45	-16.48	-17.82
Model-Informed LISTA	$\mathbf{-13.34}$	$\mathbf{-14.71}$	$\mathbf{-14.92}$	$\mathbf{-14.93}$

TABLE VIII
ALGORITHM–HARDWARE TRADEOFF (ZCU104, 100 MHZ, HLS ESTIMATES FOR K=12/16-CNN32, POST-ROUTE FOR PROPOSED)

Configuration	NMSE (20 dB)	DSP	BRAM	LUT	Latency
$K = 12$, CNN-32	-13.8 dB	650	58%	48%	19.5 ms
$K = 16$, CNN-32	-14.5 dB	650	60%	52%	22.1 ms
Proposed ($K = 16$, CNN-48)	-14.9 dB	981	64%	58%	23.3 ms

The classical non-sparse baseline models have a plateau in performance from approximately -1 dB to -5 dB, as they are limited by the null-space of Φ. this work model achieves saturation at approximately -14.9 dB after achieving an SNR greater than 10 dB. After this point, the majority of the remaining error is due to network capacity and model mismatch, rather than channel noise. For SNR levels above 20 dB, the reference method based on the sample-covariance LMMSE filter begins to perform better than this work method. This should be expected, as this method calculates an unconstrained linear MMSE filter from the true empirical covariance of all data in the training set. However, for SNR levels in the low-to-medium range (-10 dB to +15 dB), Model-Informed LISTA maintains a performance advantage of approximately 2 to 7.5 dB.

B. Ablation Study and Impact of Design Choices

The Table IX, shows how much every design decision has added. The largest individual addition was made by replacing the DFT matrix with the binary Rademacher pilot schedule (+1.87 dB). Replacing the DFT matrix with the binary Rademacher pilot schedule had no hardware cost. The addition of a residual skip connection (+0.96 dB) tells us that the CNN performs better as a residual error corrector. Increasing the depth of the core from K=12 to K=16 added +0.70 dB. Eliminating an auxiliary L1 sparsity penalty when training the model adds up +0.40 dB. Sparsity penalty appears to throw away weak physical paths.

The dynamic ENE scaling has given a constant improvement of 1.8 to 2.8 dB throughout the work . Table X shows that when using fixed thresholds for medium SNR, the model performs poor adn there is a significant change form -10 dB compared with -12.8 dB at low SNR.At high SNR, due to the fixed thresholding the model performance weak and there is a much increase in the performance when we change to ENE(-12.3 dB

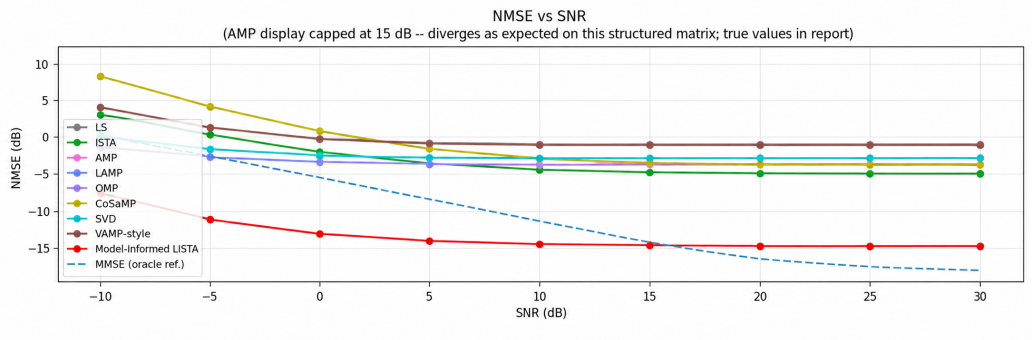


FIG. 4. NMSE versus SNR on DeepMIMO O1_28 across eight baselines, the sample-covariance linear reference, and Model-Informed LISTA. The divergent AMP curve is clipped to +15 dB for readability.

TABLE IX
ABLATION: SEQUENTIAL NMSE IMPROVEMENTS

#	Reference configuration	Modified configuration	ΔNMSE (dB)
1	sparsity penalty used	sparsity penalty removed	+0.40
2	CNN width 32	CNN width 48	+0.36
3	no skip connection	skip connection	+0.96
4	$K = 12$ layers	$K = 16$ layers	+0.70
5	DFT pilot matrix	Rademacher pilot matrix	+1.87

TABLE X
ENE ABLATION: FIXED VERSUS ENE-CONDITIONED THRESHOLDS (NMSE, dB)

Thresholds	−10 dB	0 dB	10 dB	20 dB	30 dB
Fixed	-10.0	-11.5	-12.1	-12.3	-12.3
ENE-conditioned	-12.8	-13.3	-14.7	-14.9	-14.9
Improvement	+2.8	+1.8	+2.6	+2.6	+2.6

vs. -14.9 dB), so this concludes using ENE help the model to perform better at any snr range.

C. Noise Adaptivity and Cost

Deep estimators are commonly unstable due to SNR mismatch. A model trained at 20 dB will perform poorly at 0 dB. Therefore, designers must maintain multiple networks. On the other hand, the ENE eliminates the issue of SNR mismatch. The ENE continuously adjusts its shrinkage; consequently, one 0.92 MB parameters network remains competitive over a 40 dB dynamic range.

VIII. HARDWARE RESULTS

A. Quantization Fidelity

Table XI shows how changing the bit width affects the accuracy of the results of teh work . If width drops below 16 bits then it could affect the model accuracy and also will notice a difference in quality. 14 bit math results in an average loss of 0.32dB in NMSE and 12 bit math results in an average loss of 1.45dB in NMSE. If go up to 18 bits, can only gain approximately.013dB, however, will need to increase the

TABLE XI
FIXED-POINT BITWIDTH VS. ACCURACY TRADEOFF (PROPOSED ARCHITECTURE. 12, 14, 18-BIT: HLS ESTIMATES; 16-BIT: POST-ROUTE)

Format (W, A, Acc)	NMSE Degradation	DSP	BRAM	LUT
12-bit (12, 12, 24)	+1.45 dB	840	48%	42%
14-bit (14, 14, 28)	+0.32 dB	910	55%	49%
16-bit (16, 16, 32)	**+0.065 dB**	**981**	**64%**	**58%**
18-bit (18, 18, 36)	+0.052 dB	1240	72%	69%

TABLE XII
FIXED-POINT HLS VERSUS FLOAT32: NMSE DEVIATION

Statistic	Deviation (dB)	Extra error energy
Mean absolute	0.065	+1.5%
99th percentile	0.156	+3.7%
Maximum per test	0.198	+4.7%

number of slices used on the DSP by 259 (from 981 to 1240). Therefore, the 16-bit format (`ap_fixed`) represents the point of diminishing returns.

Table XII provides a summary of results. On average, the 16-bit fixed point bit stream after routing has a difference of 0.065 dB from 32-bit floating point execution (an additional 1.5% energy loss due to quantization). At the 99th percentile, the difference remains less than 0.16 dB.

B. Resources, Timing, and Power Consumption

Table XIII presents the post-route resource utilization of the Zynq UltraScale+ XCZU7EV. The entire design utilizes 133,071 LUTs (57.8%), 80,581 flip flops (17.5%), 199.5 BRAM blocks (63.9%) and 981 DSP slices (56.8%). The timing was met at 100MHz and had a worst negative slack (WNS) of +0.210ns, representing a $f_{\max}$ of 102.1MHz.

Table XIII block-level numbers reinforce the core idea of the system design. No DSP slices are used within the projection blocks of the Φ and Φ^H by using a binary Rademacher sensing matrix; no DSPs are implemented within those projection blocks. Most remaining DSPs are utilized in the multi-channel residual CNN (821), the 2D FFT/IFFT cores (128), and the ENE (32).

TABLE XIII
RESOURCES AND LATENCY BREAKDOWN PER BLOCK (ZCU104, 100 MHZ)

Block	LUT	FF	BRAM	DSP	Cumulative Latency	Notes
Φ multiplication	9345	4500	0	0	1.5 ms	binary sign / add-subtract mapping
Φ^H multiplication	9345	4500	0	0	1.5 ms	binary sign / add-subtract mapping
FFT/IFFT	15400	8200	30	128	8.0 ms	buffered Radix-2
Soft threshold	3400	2100	0	0	1.0 ms	magnitude threshold
ENE	8500	4200	5	32	0.5 ms	2 FC layers
Residual CNN	38000	23000	100	821	7.603 ms	multiplier-heavy block
Control / DMA	2340	1200	2	0	0.342 ms	PS/PL orchestration
Interconnect & Glue Logic	46741	32881	62.5	0	2.825 ms	AXI crossbars, buffers, state machines
Total, HLS estimate	57.76%	17.49%	63.94%	56.77%	23.270 ms	unrolled core stage: 15.325 ms
Total, post-route	133071	80581	199.5	981	—	$f_{\max}$ 102.1 MHz, WNS 0.210ns / TNS 0.0ns

TABLE XIV
ON-BOARD TEST CAMPAIGNS ON THE ZCU104

Set	Reference	Pass if	Passed	Avg NMSE
V1 (10)	golden model	< -25 dB	10/10	-43.43 dB
V2 (10)	DeepMIMO truth	< -10 dB	10/10	-18.73 dB
V2 (1000)	DeepMIMO truth	≤ -10 dB	933/1000	-16.96 dB

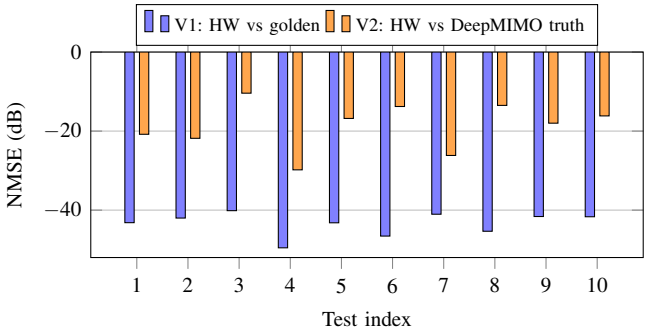


FIG. 5. On-board NMSE for V1 (software agreement) and V2 (ray-traced channel accuracy).

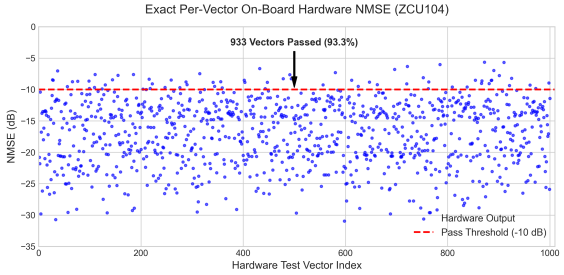


FIG. 6. On-board NMSE scatter across 1000 test cases: mean NMSE -16.96 dB, 933 passed (≤ -10 dB), 67 failed, mean latency 23.270 ms.

TABLE XV
FAILURE-TAIL CHARACTERISTICS (≤ -10 DB PASS VS. FAIL)

Property	Passed (933)	Failed (67)
Dominant paths (mean)	6.2 ± 2.1	18.4 ± 4.3
Rel. channel energy (dB)	0.0 ± 1.2	-8.5 ± 2.4
User position	Central grid	Grid edges
Float32 NMSE (mean)	-17.02 dB	-9.21 dB
Fixed-pt NMSE (mean)	-16.95 dB	-9.14 dB

Power Consumption: The total on-chip power as estimated by the Vivado Power Analyzer based on post-route switching activity during typical operations is 1.3W for both the PL and PS. Total energy used over a 23.27ms execution period is approximately $1.3\text{W} \times 23.27\text{ms} \approx 30.3$ mJ per channel estimate.

C. On-Board Validation

Table XIV and Figs. 5 and 6 illustrate this work findings on the actual physical board. For campaign V1 (FPGA hardware versus golden model), the FPGA averaged -43.43 dB NMSE across 10 vectors, ranging from -49.54 dB to -40.14 dB. The RTL execution and the software model are bit-accurately matched.

All ten initial vectors are found to be below -10dB for the campaign V2 hardware vs. true channels. The average normalized mean square error (NMSE) for these vectors was -18.73dB. During the entire run of the 1000-vector campaign, the board achieved an average of -16.96dB, and 933 of the 1000 vectors (93.3%) fell below or equal to -10dB (95% Wilson confidence interval: 91.6%–94.7%).

D. Characterization of the Failure Tail

The -10dB threshold was not met by sixty-seven channels (6.7%) in hardware. Table XV demonstrates that these channels have different structural characteristics than those which met the -10dB threshold. On average, the channels that did not meet the -10dB threshold had 18.4 ± 4.3 dominant multipath components. On average, the channels that did meet the -10 dB threshold had only 6.2 ± 2.1. Since compression recovery depends upon angular sparsity, having nearly three times as many active paths negatively impacts the recovery process.

The failed channels have approximately 8.5dB less total channel energy than the other channels which indicates that they are suffering from deep fading. Many of the users with

TABLE XVI
MEASURED ON-BOARD LATENCY (MEAN, ms)

Stage	10 runs	1000 runs	Share
Unrolled core	15.325	15.325	65.9%
Refinement CNN	7.603–7.604	7.603	32.7%
Other (PS steps)	0.336–0.337	0.342	1.5%
Total	**23.264–23.265**	**23.270**	100%
Std. / min / max	0.002 ms	23.265 ms	23.275 ms

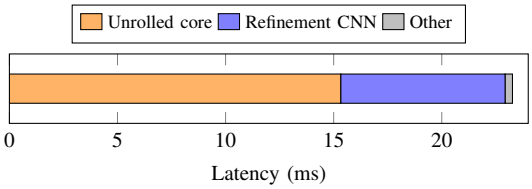


FIG. 7. Latency breakdown across the 23.270 ms execution time (1000-run campaign).

failed channels are located at the edge of the DeepMIMO grid. At the edges of the DeepMIMO grid, the steering angle for the RIS is very large. This is perhaps the most important point. The fixed-point NMSE on the failed channels (-9.14dB) is within 0.07dB of the 32-bit floating-point value (-9.21dB). Thus, the failures are due to the channel geometry and limitations on the number of non-zero elements in the channel matrix and not due to the errors introduced by the fixed-point implementation.

E. Latency

Table XVI and Fig. 7 present the end-to-end latency data that was collected. For all 1000 trials, the average was 23.270ms; the standard deviation of this data set was 0.002ms. The trial which resulted in the smallest end-to-end latency had a value of 23.265ms while the trial with the largest had a value of 23.275ms.

The sixteen unrolled layers of LISTA consume 15.325 ms, representing approximately 65.9 percent of the total time, or 1.53 million clock cycles (at 100 MHz). The remainder of the processing time, 7.603 ms (about 32.7 percent of the total time; about 760,000 clock cycles) is used by the residual CNN. In addition to these two components, software orchestration on the Cortex-A53 including cache invalidation, memory transposition, and re-arming the DMA represents another 0.342 ms (about 1.5 percent) of processing time.

IX. COMPARING THIS WORK WITH THE MOST SIMILAR ZYNQ IMPLEMENTATION

This work is placed within the context of the embedded communication literature through Table XVII.

Where Model-Informed LISTA excels: The design of Model-Informed LISTA addresses a far more challenging mathematical issue. Model-Informed LISTA recovers 1040 spatial parameters based on only 256 compressed measurements ($M/N \approx 0.246$). In contrast, LSiDNN can interpolate 48 pilot subcarriers to eliminate 864 ris elements using typical statistical tap models, and even plain interpolation provides a reasonably good starting point. Additionally, LSiDNN suffers when there is an SNR mismatch between the training data and testing data; therefore, LSiDNN must load multiple models from memory.But the present model ENE dynamically adjusts its threshold values across the entire 40dB dynamic range. Finally, this work has been verified using ray-traced ground truth during more than 1000 hardware runs, resulting in only 0.065dB of quantization error.

X. DISCUSSION AND LIMITATIONS

This work can be used in hardware because of its low end-to-end latency of 23.27 ms, and low use of DSPs (56.8%). It also provides insight into the areas for improvement. Memory movement between kernels is identified as the biggest performance limitation through profiling. There are three potential methods to reduce latency:

1) Move the transpose of the matrix into the programmable logic (PL) using a streaming line buffer; thus, preventing data from traveling to and from the Cortex-A53 processor.
2) Pipeline each of the 16 unrolled layers, such that every subsequent estimate of the channels is processed sequentially by each of the 16 layers.
3) Replace the current AXI DMA interrupt scheme with scatter gather DMA and prevent the reset per-estimate driver.

The study has some limitations:

1) *The Single Propagation Case:* All performance evaluations used the DeepMIMO O1_28 Urban Outdoor Scenario. The logical follow-up evaluation will be the assessment of the proposed system performance in other types of environments (i.e., indoor), and under more complex motion patterns.
2) *Power reporting:* Power results report are based on post-route Vivado switching activity analysis, rather than measurement of current flowing through shunt resistors placed on the board's power rails.

Practical Implications: The 23 ms latency was within the channel coherence time limit for both pedestrian and quasi-static environments, which have channel coherence times greater than 50-100 ms. However, the latency was too high for vehicular communication systems. Fixed point measurements indicated that algorithm unrolling would allow for a low cost of reducing the word length as the physical sensing matrix is kept and only the scalar step size is trained. Additionally, pilot-matrix selection resulted in a higher NMSE improvement (+1.87 dB) than increasing the number of layers in the network. Therefore, there is a practical implication that when designing the surface phase patterns and baseband reconstruction algorithm they should be developed simultaneously.

TABLE XVII
DIMENSION-BY-DIMENSION COMPARISON WITH LSiDNN. VALUES FOR LSiDNN ARE AS REPORTED THERE.

Dimension	LSiDNN	Model-Informed LISTA (this work)
Task	Pilot-based OFDM channel estimation (LS at 48 pilots, interpolated to 864 REs)	Passive-RIS cascaded-channel *sparse recovery* from compressed measurements ($M/N \approx 0.246$)
Channel data	MATLAB 3GPP statistical models (EPA/EVA/ETU), 10,000 frames per channel	DeepMIMO ray-traced O1_28 scenario
Model	3-layer fully connected net, $96 \to 48 \to 1728$; 103,440 params, 0.1 M MACs	16-layer model-informed unrolled core + ENE + CNN; 230,749 params
Model structure	Black-box FCNN	Interpretable: layers follow ISTA iterations with the known sensing operator
Noise/SNR handling	Single network trained for given channel conditions; accuracy degrades under train/test mismatch; handled by memory-based model swapping	ENE conditions thresholds on SNR; one model across -10 to 30 dB
Quantization	Post-hoc word-length search; $(26, 8)$ for LSiDNN	Post-training fixed-point conversion; 0.065 dB mean deviation from float32
Platform	ZC706 (Zynq-7000, Cortex-A9)	ZCU104 (Zynq UltraScale+ XCZU7EV, Cortex-A53)
Baselines	LS, LMMSE, iResNet	LS, ISTA, AMP, LAMP, OMP, CoSaMP, SVD, VAMP-style, LMMSE-style reference
Hardware validation	MSE/BER from simulation; execution time, resthis workces, power on board	HLS C/RTL co-sim (20 tests); 1000 on-board vectors scored against ground truth
Downstream metric	BER	NMSE
Execution time	0.086 μs (LL) / 0.715 μs (CE)	23.27 ms end-to-end (system level, highly repeatable)
Resthis workces	LL: 234 BRAM, 32 DSP; CE: 8 BRAM, 6 DSP	63.9% BRAM, 56.8% DSP (HLS est.)

XI. CONCLUSION

An algorithm-hardware co-design approach has been proposed for a model-based deep-unfolded channel estimator in passive RIS systems. The model is based on a binary Rademacher sensing matrix, a 16-layer unrolled ISTA core, ENE threshold conditioning, and residual CNN refinement. This method provides NMSE improvement of 13.7dB when compared to least squares (LS) and 9.6dB when compared to iterative ISTA. Custom C++ HLS kernels have been developed to implement the algorithm with 16-bit fixed point arithmetic, and the performance gap with respect to 32-bit floating point arithmetic is reported to be within 0.065dB. The accelerator was implemented on an AMD/Xilinx ZCU104 platform at 100MHz. This model achieved a mean NMSE of –16.96dB across 1,000 ray-traced test channels, a pass rate of 93.3% and a latency of 23.27ms. Future directions include streaming data onto the chip directly to minimize processor round trip times, multi-user scheduling extensions to the current design, and board-level power measurements.

APPENDIX A
ERROR PROPAGATION THROUGH UNROLLED LAYERS

Layer k of the unrolled core applies the operator

$$T_k(\mathbf{h}) = \widetilde{\mathbf{F}}^H \eta_{\theta_k}\left(\widetilde{\mathbf{F}}\left[(\mathbf{I} - \gamma_k \Phi^H \Phi)\mathbf{h} + \gamma_k \Phi^H \mathbf{y}\right]\right). \tag{9}$$

The unitary transform $\widetilde{\mathbf{F}}$ preserves Euclidean norms ($\|\widetilde{\mathbf{F}}\mathbf{x}\|_2 = \|\mathbf{x}\|_2$). The soft-thresholding operator η_θ is non-expansive, that is, 1-Lipschitz:

$$\|\eta_\theta(\mathbf{u}) - \eta_\theta(\mathbf{v})\|_2 \le \|\mathbf{u} - \mathbf{v}\|_2, \quad \forall \mathbf{u}, \mathbf{v} \in \mathbb{C}^N. \tag{10}$$

For the linear step, the spectral norm is $\|\mathbf{I} - \gamma_k \Phi^H \Phi\|_2 = \max_i |1 - \gamma_k s_i^2|$, where s_i are the singular values of Φ. If the learned step size obeys $0 < \gamma_k \le 2/\|\Phi\|_2^2$, this norm is bounded by 1, and every unrolled stage is non-expansive.

Let δ_k be the additive quantization noise that 16-bit fixed-point arithmetic introduces at layer k. The error vector $\mathbf{e}^{(k+1)} = \hat{\mathbf{h}}^{(k+1)} - \mathbf{h}^{(k+1)}$ then satisfies

$$\|\mathbf{e}^{(k+1)}\|_2 \le \|\mathbf{e}^{(k)}\|_2 + \|\delta_k\|_2. \tag{11}$$

Applying the recurrence over all K layers gives

$$\|\mathbf{e}^{(K)}\|_2 \le \sum_{k=0}^{K-1} \|\delta_k\|_2. \tag{12}$$

Quantization error thus grows at most linearly with depth. If model rounding as an independent uniform error on each of the $2N$ real entries, get $\mathbb{E}[\|\delta_k\|_2^2] \approx 2N\Delta^2/12$. This predicts a conservative theoretical noise floor of $K \cdot 2N\Delta^2/(12\|\mathbf{h}\|_2^2)$. The degradation measure is only 0.065 dB, so 16-bit words preserve floating-point convergence in practice.

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